

● RAG Learning

3.2 Percentages, Growth & Decay

AQA GCSE Mathematics 8300 — Ratio, Proportion & Rates of Change — spec refs R9, R16 — Papers 1 (non-calculator), 2 and 3 (calculator)

3.2.1 What a Percentage Means

Per cent means “number of parts per hundred”. $35\% = \frac{35}{100} = 0.35$.

A percentage can be **greater than 100%**: 120% of an amount is 1.2 times the amount.

Percentage	Fraction	Decimal
50%	$\frac{1}{2}$	0.5
25%	$\frac{1}{4}$	0.25
37.5%	$\frac{3}{8}$	0.375
4.5%	$\frac{9}{200}$	0.045
150%	$\frac{3}{2}$	1.5

Percentages of an amount without a calculator

Build up from 10% (divide by 10), 5% (half of 10%), 1% (divide by 100) and so on.

Worked Example: Work out 17.5% of £360.

$10\% = 36$, $5\% = 18$, $2.5\% = 9$. So $17.5\% = 36 + 18 + 9 = \text{£}63$.

One quantity as a percentage of another

A as a percentage of $B = \frac{A}{B} \times 100\%$ (same units first)

Worked Example: (a) 45 minutes as a percentage of 2 hours: $\frac{45}{120} \times 100 = \mathbf{37.5\%}$.
 (b) £18 as a percentage of £15: $\frac{18}{15} \times 100 = \mathbf{120\%}$.

Comparing using percentages

Worked Example: Sam scored 42 out of 60 in test A and 54 out of 75 in test B. Which is the better result?

Test A: $\frac{42}{60} = 70\%$. Test B: $\frac{54}{75} = 72\%$. **Test B** is better.

3.2.2 Percentage Increase and Decrease with Multipliers

Interpret a percentage change **multiplicatively**: the new amount is a percentage of the original, written as a decimal **multiplier**.

Increase by $r\%$: multiplier = $1 + \frac{r}{100}$ e.g. $+12\% \rightarrow 1.12$

Decrease by $r\%$: multiplier = $1 - \frac{r}{100}$ e.g. $-18\% \rightarrow 0.82$

New amount = original $\times$ multiplier

original = 100% 12% $\times 1.12$

82% remains -18% $\times 0.82$

Increase: multiplier above 1. Decrease: multiplier below 1.

Worked Example: (a) Increase £85 by 12%: $85 \times 1.12 = \text{£}95.20$.
 (b) A TV costing £450 is reduced by 18%: $450 \times 0.82 = \text{£}369$.

Percentage change

$$\text{Percentage change} = \frac{\text{change}}{\text{original}} \times 100\%$$

Worked Example: A ticket price rises from £36 to £45. Change = 9. Percentage increase
 $= \frac{9}{36} \times 100 = 25\%$.

A dog's mass falls from 80 kg to 68 kg. Percentage decrease $= \frac{12}{80} \times 100 = 15\%$.

Exam Tip: Always divide by the **original** value, not the new one. Dividing 12 by 68 gives 17.6%, a common wrong answer.

Profit and loss

Percentage profit $= \frac{\text{profit}}{\text{cost price}} \times 100$. A trader who buys 40 items at £6 (cost £240) and sells 30 at £10 and 10 at £4 (income £340) makes £100 profit: $\frac{100}{240} \times 100 = 41.7\%$.

3.2.3 Reverse Percentages (Original Value Problems)

In a **reverse percentage** problem you know the amount *after* a percentage change and want the original. Since $\text{new} = \text{original} \times \text{multiplier}$:

$$\text{original} = \frac{\text{new amount}}{\text{multiplier}}$$

Worked Example: After a 15% pay rise, Jo earns £29 900. What did she earn before?

$$115\% = 29\,900, \text{ so original} = 29\,900 \div 1.15 = \textbf{£26\,000}.$$

Worked Example: A coat costs £52 after a 35% reduction. Find the original price.

$$65\% = 52, \text{ so original} = 52 \div 0.65 = \textbf{£80}.$$

Worked Example: A price of £66 includes VAT at 20%. Price before VAT = $66 \div 1.2 = \textbf{£55}$.

Exam Tip: Never take the percentage of the *new* amount and add or subtract it. 20% of £66 is £13.20, and $£66 - £13.20 = £52.80$ is wrong — the 20% was of the original price.

Successive percentage changes

Multiply the multipliers to get one overall multiplier.

Worked Example: A bike's price is increased by 10% and then decreased by 10%.

Overall multiplier = $1.1 \times 0.9 = 0.99$, so the final price is 99% of the original: a **1% decrease**, not "no change".

Worked Example: Prices are cut by 15%, then a further 10% is taken off the sale price.

$$0.85 \times 0.9 = 0.765: \text{ total reduction} = 100\% - 76.5\% = \textbf{23.5\%} \text{ (not 25\%)}.$$

3.2.4 Simple Interest

Simple interest is paid only on the original amount (the principal), so the same interest is earned every year.

$$\text{Simple interest} = P \times \frac{r}{100} \times n$$

P = principal, r = rate per year (%), n = number of years

Worked Example: £2400 is invested for 3 years at 3.5% per year simple interest.

Interest per year = $2400 \times 0.035 = \text{£}84$. Total interest = $84 \times 3 = \text{£}252$.

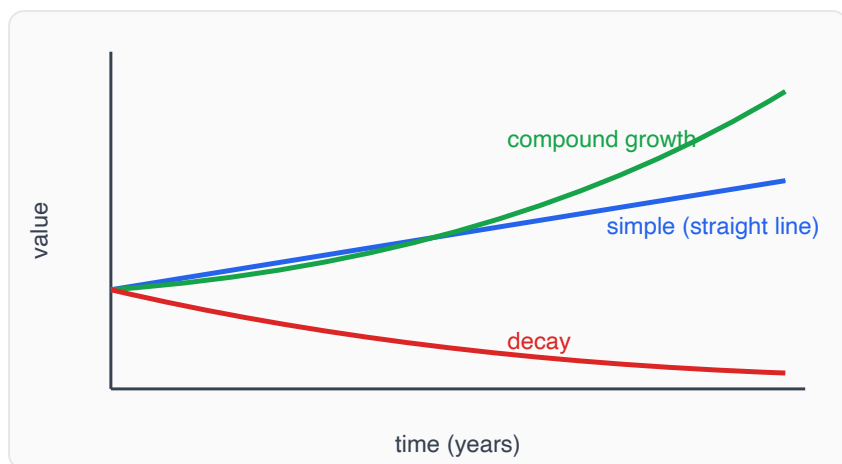
Value at the end = $2400 + 252 = \text{£}2652$.

3.2.5 Compound Interest, Growth and Decay

Compound interest: interest is added to the account each year and then also earns interest. **Growth** and **decay** (e.g. populations, depreciation of cars) work the same way — the same percentage change is applied repeatedly.

$$\text{Amount after } n \text{ periods} = P \times \left(1 \pm \frac{r}{100}\right)^n$$

Use + for growth/interest and – for decay/depreciation.



Worked Example: £6000 is invested at 2% per year compound interest for 3 years.

$$6000 \times 1.02^3 = 6367.248 \approx \textbf{£6367.25}.$$

Worked Example: A car bought for £18 500 depreciates by 12% each year. Find its value after 4 years.

$$\text{Multiplier} = 0.88. \text{ Value} = 18\,500 \times 0.88^4 = 11\,094.36 \dots \approx \textbf{£11\,094}.$$

Worked Example: A van costs £24 000. It loses 20% in the first year and 15% in each year after that. Value after 3 years:

$$24\,000 \times 0.8 \times 0.85^2 = \textbf{£13\,872}.$$

Comparing simple and compound interest

Worked Example: Bank A: £5000 at 4% simple interest for 3 years. Bank B: £5000 at 3.9% compound interest for 3 years.

$$\text{A: } 5000 + 3 \times 200 = \textbf{£5600}. \quad \text{B: } 5000 \times 1.039^3 = \textbf{£5608.11}.$$

Bank B gives more, even though its rate is lower, because the interest compounds.

Finding the number of years

Use trial and improvement with the formula: test whole numbers of years and show the values either side of the target.

Worked Example: A lake has 1200 fish, decreasing by 8% a year. When will there first be fewer than 800?

$$1200 \times 0.92^4 = 859.7 \text{ (still above 800); } 1200 \times 0.92^5 = 790.9 \text{ (below 800).}$$

Answer: after **5 years**.

Reverse growth and finding the rate

Worked Example: After 2 years of 10% compound growth an investment is worth £4840. Original = $4840 \div 1.1^2 = 4840 \div 1.21 = \text{£}4000$.

Worked Example: £8000 grows to £9724.05 in 4 years at $r\%$ compound interest. Find r .

$$1 + \frac{r}{100} = \sqrt[4]{\frac{9724.05}{8000}} = \sqrt[4]{1.21550625} = 1.05, \text{ so } r = 5.$$

Exam Tip: "Compound", "each year" or "per annum, depreciates" means use a power. "Simple" means the same interest every year. Read whether the question wants the *total value* or just the *interest*.

3.2.6 General Iterative Processes Higher

Many real situations combine a percentage change with a fixed amount added or removed each period. These are described by an **iterative formula** (a recurrence relation):

$$u_{n+1} = a u_n + b$$

u_0 is the starting value, a is the multiplier (percentage change), b is the fixed amount added ($b > 0$) or removed ($b < 0$) each period.

Formula	Meaning
$u_{n+1} = 1.05u_n - 20$	grows by 5% each period, then 20 is removed
$u_{n+1} = 1.03u_n + 100$	3% interest added each year, then £100 deposited
$P_{n+1} = 0.9P_n + 50$	10% of the population is lost, then 50 are added
$B_{n+1} = 1.01B_n - 300$	loan with 1% monthly interest and £300 repaid each month

Worked Example: $u_{n+1} = 1.05u_n - 20$ and $u_0 = 500$. Find u_2 .

$$u_1 = 1.05 \times 500 - 20 = 505$$

$$u_2 = 1.05 \times 505 - 20 = 530.25 - 20 = \mathbf{510.25}$$

Worked Example: Island population: $P_{n+1} = 1.2P_n - 150$, $P_0 = 1000$ in 2020. When does it first exceed 1500?

$$P_1 = 1050, P_2 = 1110, P_3 = 1182, P_4 = 1268.4, P_5 = 1372.08, P_6 = 1496.50, P_7 = 1645.80$$

First exceeds 1500 at $n = 7$, i.e. in **2027**.

Long-term behaviour

If the sequence settles down, the limit L satisfies $L = aL + b$.

Worked Example: $u_{n+1} = 0.8u_n + 30$. Long-term value: $L = 0.8L + 30 \Rightarrow 0.2L =$

$$30 \Rightarrow L = 150.$$

Worked Example: A £5000 loan at 1% per month with only £50 repaid each month: $B_1 = 1.01 \times 5000 - 50 = 5000$. The interest (£50) exactly cancels the repayment, so the balance never falls — the loan is never repaid.

Exam Tip: Use your calculator's ANS key: type the starting value, press =, then type $1.05 \times \text{ANS} - 20$ and press = repeatedly. Write down each term to show your method, and keep full accuracy until the end.

Summary

Problem type	Method
Percentage of an amount	Amount $\times$ decimal (or build up from 10%, 5%, 1%)
A as a % of B	$\frac{A}{B} \times 100$
Increase / decrease	Original $\times$ multiplier
Percentage change	$\frac{\text{change}}{\text{original}} \times 100$
Original value	New $\div$ multiplier
Simple interest	$P \times \frac{r}{100} \times n$
Compound growth / decay	$P \times \text{multiplier}^n$
Iterative process Higher	$u_{n+1} = au_n + b$, apply repeatedly